

Primer on Bounded and Compact Operators in Banach and Hilbert Spaces

The primary focus of this course will be the finite element approximation of the spectra of compact operators in Banach spaces.

From this moment on we will denote by $(X, \|\cdot\|_X)$ and $(Y, \|\cdot\|_Y)$ two Banach spaces with the corresponding norm. Furthermore, we will denote by $\mathcal{L}(X, Y)$ the set of linear operators from X to Y .

Definition 0.1. Let $T \in \mathcal{L}(X, Y)$. We say that T is a bounded operator if there exists $C_T > 0$ such that, for all $x \in X$,

$$\|Tx\|_Y \leq C_T \|x\|_X.$$

We will denote the set of all bounded linear operators by $\mathcal{B}(X, Y)$.

The space of bounded linear operators equipped with the norm

$$\|T\|_{\mathcal{B}(X, Y)} := \sup_{x \in X} \frac{\|Tx\|_Y}{\|x\|_X}$$

is a Banach space.

Theorem 0.2. Let $T \in \mathcal{B}(X, Y)$ be a bijective operator. Then $T^{-1} \in \mathcal{B}(Y, X)$. In particular,

$$\|T^{-1}\|_{\mathcal{B}(Y, X)} \geq \|T\|_{\mathcal{B}(X, Y)}^{-1}.$$

Proof. This result is a consequence of the open mapping theorem. □

Theorem 0.3 (First stability estimate). Let $T \in \mathcal{B}(X, Y)$ be a bijective operator. Then for every $\Delta \in \mathcal{B}(X, Y)$ such that

$$\|\Delta\|_{\mathcal{B}(X, Y)} < \|T^{-1}\|_{\mathcal{B}(Y, X)}^{-1},$$

the operator $T + \Delta$ is bijective, $(T + \Delta)^{-1} \in \mathcal{B}(Y, X)$, and the following bound holds:

$$\|(T + \Delta)^{-1}\|_{\mathcal{B}(Y, X)} \leq \|T^{-1}\|_{\mathcal{B}(Y, X)} \left[1 - \|\Delta\|_{\mathcal{B}(X, Y)} \|T^{-1}\|_{\mathcal{B}(Y, X)} \right]^{-1}. \quad (*)$$

Proof. Consider

$$X \xrightarrow{\Delta} Y \xrightarrow{T^{-1}} X.$$

We begin by observing that

$$\|\Delta T^{-1}\|_{\mathcal{B}(X, X)} \leq \|\Delta\|_{\mathcal{B}(X, Y)} \|T^{-1}\|_{\mathcal{B}(Y, X)}.$$

Hence the Neumann series

$$\sum_{k=0}^{\infty} (-\Delta T^{-1})^k$$

converges. In particular, the Neumann series converges to

$$(I_X + \Delta T^{-1})^{-1},$$

and

$$\|(I_X + \Delta T^{-1})^{-1}\|_{\mathcal{B}(X,X)} \leq \sum_{k=0}^{\infty} \|(-\Delta T^{-1})^k\|_{\mathcal{B}(X,X)} \leq \frac{1}{1 - \|\Delta T^{-1}\|_{\mathcal{B}(X,X)}}.$$

Here we used

$$\sum_{k=0}^{\infty} ar^k = \frac{a}{1-r}, \quad |r| < 1.$$

Notice now that

$$T + \Delta = (I_X + \Delta T^{-1})T.$$

Since both T and $(I_X + \Delta T^{-1})$ are invertible, $T + \Delta$ is invertible, and

$$(T + \Delta)^{-1} = T^{-1}(I_X + \Delta T^{-1})^{-1}.$$

Therefore

$$\|(T + \Delta)^{-1}\| \leq \|T^{-1}\|_{\mathcal{B}(Y,X)} \left(1 - \|\Delta\|_{\mathcal{B}(X,Y)} \|T^{-1}\|_{\mathcal{B}(Y,X)}\right)^{-1}.$$

□

It is now time to introduce the notion of adjoint of $T \in \mathcal{B}(X, Y)$.

From this moment on, we will denote by X^* the dual space to X , i.e. $\mathcal{B}(X, \mathbb{R})$, and we will write

$$\langle x_*, x \rangle_{X^*, X}$$

to denote the action of the element $x_* \in \mathcal{B}(X, \mathbb{R})$ on the element $x \in X$.

We now deal with the construction of an adjoint operator. As we consider $y_* \in Y^*$, we can introduce the operator

$$x \longmapsto \langle y_*, Tx \rangle_{Y^*, Y}.$$

Notice that this operator belongs to X^* , since

$$|\langle y_*, Tx \rangle_{Y^*, Y}| \leq \|y_*\|_{\mathcal{B}(Y, \mathbb{R})} \|Tx\|_Y \leq \|T\|_{\mathcal{B}(X, Y)} \|y_*\|_{\mathcal{B}(Y, \mathbb{R})} \|x\|_X.$$

Lemma 0.4 (Riesz). *Let $A \subsetneq X$ be a closed subspace. Then for every $\varepsilon > 0$, there exists $x \in X$ such that*

$$\|x\|_X = 1, \quad \text{dist}(x, A) \geq 1 - \varepsilon.$$

Definition 0.5. Given an operator $T \in \mathcal{B}(X, Y)$, we define its adjoint as the operator

$$T^* \in \mathcal{B}(Y^*, X^*)$$

such that

$$T^* : Y^* \rightarrow X^*, \quad y_* \longmapsto G_{y_*} \in \mathcal{B}(X, \mathbb{R}),$$

where G_{y_*} is defined by

$$G_{y_*}(x) = \langle y_*, Tx \rangle_{Y^*, Y}.$$

If $(X, \|\cdot\|_X)$ is a Hilbert space, many more things can be said about such operators. To make the exposition clearer, we will denote by $(H, \|\cdot\|_H)$ and $(K, \|\cdot\|_K)$ two Hilbert spaces, and by $(\cdot, \cdot)_H$ and $(\cdot, \cdot)_K$ the respective inner products.

We will make heavy use of the Riesz–Frechet identification.

Theorem 0.6 (Riesz–Frechet identification). *Given any $h_* \in H^*$, there exists a unique $h \in H$ such that*

$$\langle h_*, k \rangle_{H^*, H} = (h, k)_H$$

for all $k \in H$. Furthermore,

$$\|h\|_H = \|h_*\|_{H^*}.$$

We call the Riesz map of H

$$R_H : H^* \rightarrow H, \quad h_* \mapsto h.$$

Then R_H is bijective.

Proof. The proof of this result can be found in Brezis, Theorem 5.5. □

The adjoint operator in the Hilbert setting can be defined starting from the Banach adjoint as follows.

Definition 0.7. Given $T \in \mathcal{B}(H, K)$ and its Banach adjoint $T^* \in \mathcal{B}(K^*, H^*)$, we define the Hilbert adjoint of T to be the operator

$$T' : K \rightarrow H, \quad k \mapsto R_H T^* R_K^{-1} k.$$

Proposition 0.8. *Let $T \in \mathcal{B}(H, K)$. Then the Hilbert adjoint T' can also be defined as the operator $T' \in \mathcal{B}(K, H)$ such that, for all $(h, k) \in H \times K$,*

$$(Th, k)_K = (h, T'k)_H. \tag{*}$$

Proof. We compute

$$\begin{aligned} (h, T'k)_H &= (h, R_H T^* R_K^{-1} k)_H \\ &= \langle T^* R_K^{-1} k, h \rangle_{H^*, H} \\ &= \langle G_{R_K^{-1} k}, h \rangle_{H^*, H} \\ &= \langle R_K^{-1} k, Th \rangle_{K^*, K} \\ &= (k, Th)_K. \end{aligned}$$

Thus we have the identity

$$(Th, k)_K = (h, T'k)_H.$$

□

We can now define two more categories of endomorphisms, which will later have particular spectral properties.

Definition 0.9. We say that an operator $T \in \mathcal{B}(H, H)$ is self-adjoint if

$$T' = T.$$

We say that T is normal if

$$T'T = TT'.$$

Remark 0.10. It is important to notice that in the previous definition T was an endomorphism, i.e.

$$T \in \mathcal{B}(H, H).$$

Remark 0.11. We have here omitted the identification of H with H^* via the Riesz map to avoid unpleasant pseudo-nice situations such as the fact that if

$$H_0^1 \subset H^1 \quad \text{and} \quad L^2 \simeq (L^2)^*,$$

then

$$H_0^1 \subset L^2 \simeq (L^2)^* \subset H^{-1} = (H_0^1)^*.$$

Theorem 0.12. *If $T \in \mathcal{B}(H, H)$ is self-adjoint, then $T^{-1} \in \mathcal{B}(H, H)$ is self-adjoint.*

Definition 0.13. We say that an operator $T \in \mathcal{L}(X, Y)$ is compact if for every bounded sequence $\{x_n\}_{n \in \mathbb{N}} \subset X$, one can extract a converging subsequence from $\{Tx_n\}_{n \in \mathbb{N}}$ in Y . We denote the space of compact operators from X to Y by $\mathcal{K}(X, Y)$.

Corollary 0.14. *All compact operators are bounded operators, i.e.*

$$\mathcal{K}(X, Y) \subseteq \mathcal{B}(X, Y).$$

Proof. Let $T \in \mathcal{K}(X, Y)$ and assume that $T \notin \mathcal{B}(X, Y)$. Then there must exist a sequence $\{x_n\}_{n \in \mathbb{N}} \subset X$ such that

$$\|x_n\|_X \leq 1, \quad \|Tx_n\|_Y > n, \quad n \in \mathbb{N}.$$

Such a sequence has no bounded subsequence after applying T , and therefore has no Cauchy subsequence, implying $T \notin \mathcal{K}(X, Y)$. $\square$

Proposition 0.15. *Given $S \in \mathcal{B}(X, X)$ and $T \in \mathcal{K}(X, Y)$, then*

$$TS \in \mathcal{K}(X, Y).$$

Proof. Bounded operators preserve boundedness of sequences; hence TS must be compact. $\square$

Theorem 0.16 (Fredholm alternative). *Let $T \in \mathcal{K}(X, X)$. Then*

$$\mathcal{N}(T - I) = \{0\} \quad \text{if and only if} \quad \mathcal{R}(T - I) = X.$$

Furthermore, $\mathcal{N}(T - I)$ is finite-dimensional.

Proof. Omitted in the handwritten notes. $\square$

Corollary 0.17. *Let $T \in \mathcal{K}(X, X)$ and $z \in \mathbb{C} \setminus \{0\}$. Then either there are non-trivial solutions to the equation*

$$Tx = zx,$$

or $T - zI_X$ is bijective and

$$(T - zI_X)^{-1} \in \mathcal{B}(X, X).$$

Proof. Notice that if $T \in \mathcal{K}(X, X)$, then $z^{-1}T \in \mathcal{K}(X, X)$ for any $z \neq 0$. We can apply the previous theorem to state that

$$\mathcal{N}(T - zI_X) = \{0\} \quad \text{if and only if} \quad \mathcal{R}(T - zI_X) = X.$$

Hence either the operator is bijective and has bounded inverse, because $T \in \mathcal{B}(X, X)$ and we can apply the bounded inverse theorem to $T - zI$, or it has a non-trivial kernel. $\square$

Example 0.18. Let us consider the differential operator

$$T_0 : W_0^{1,1+\varepsilon}([a, b], \mathbb{R}) \rightarrow L^{1+\varepsilon}([a, b], \mathbb{R}), \quad T_0 u := u'(x).$$

We can compute the Banach adjoint

$$T_0^* : L^q([a, b], \mathbb{R}) \rightarrow W^{-1,q}([a, b], \mathbb{R}), \quad q = \frac{1+\varepsilon}{\varepsilon}.$$

Following the definition of the Banach adjoint, we have that T_0^* can be defined by its action: for all $f_* \in L^q([a, b])$,

$$\langle T_0^* f_*, u \rangle_{W^{-1,q}, W_0^{1,1+\varepsilon}} = \langle f_*, T_0 u \rangle_{L^q, L^{1+\varepsilon}}.$$

Unfortunately, this is as far as the definition of Banach adjoint takes us.

One might think the situation is different for the operator

$$T_0 : H_0^1(\Omega) \rightarrow L_0^2(\Omega),$$

yet trying to apply (*) shows that no explicit computation can be applied to obtain an expression for T' . Indeed,

$$(Tu, f)_{L^2} := \int u' f \, dx = - \int u f' \, dx \neq (u, T' f)_{H_0^1(\Omega)}.$$

Example 0.19. Let us now consider the operator

$$T_1 : W_0^{1,1+\varepsilon}([a, b]) \rightarrow W^{-1,1+\varepsilon}([a, b]), \quad u \mapsto -u'',$$

where again not much can be said of T_1 , neither in the Banach setting nor in the Hilbert setting except for the fact that

$$T_1 \in \mathcal{B}(X, Y).$$

Since $T_1 \in \mathcal{B}(X, Y)$, we play the “trick of the inverse.” In fact, we will consider the solution operator

$$T_1^{-1} : Y \rightarrow X$$

and consider such operator as

$$T_1^{-1} : L^{1+\varepsilon}([a, b]) \rightarrow L^{1+\varepsilon}([a, b]).$$

We then notice that if

$$-u'' = f,$$

then

$$- \int u'' u = \int u' u' = \int f u.$$

Using Poincaré’s inequality and Holder’s inequality, we get

$$\|u'\|_{L^2}^2 \leq \|u\|_{L^q} \|f\|_{L^{1+\varepsilon}}.$$

Now notice that by the embedding

$$W_0^{1,1+\varepsilon}([a, b]) \subset L^q([a, b]),$$

and the Poincaré inequality in the Sobolev space $W_0^{1,1+\varepsilon}([a, b])$, we reach that for every $f \in L^{1+\varepsilon}([a, b])$ we have

$$u \in W_0^{1,1+\varepsilon}([a, b]).$$

Since $W^{1,1+\varepsilon}([a, b])$ is compactly embedded in $L^{1+\varepsilon}([a, b])$, we see that

$$T_1^{-1} : L^{1+\varepsilon}([a, b]) \rightarrow L^{1+\varepsilon}([a, b])$$

is compact.

In the Hilbert setting, we can study the inverse of

$$T_1 : H_0^2([a, b]) \rightarrow L^2([a, b])$$

with extended domain

$$T_1^{-1} : L^2([a, b]) \rightarrow L^2([a, b])$$

and apply the same reasoning to see that T_1^{-1} is compact. Furthermore, let us denote

$$u = T_1^{-1}f, \quad v = T_1^{-1}g.$$

Then

$$-u'' = f, \quad -v'' = g.$$

Thus, multiplying by v and respectively by u , and integrating, we get for all $f, g \in L^2([a, b])$:

$$\begin{aligned} (f, v) &= \int_a^b -u''v \\ &= \int_a^b u'v' \\ &= - \int_a^b uv'' \\ &= (u, g). \end{aligned}$$

Thus T_1^{-1} is self-adjoint. By elliptic regularity, we know that

$$\mathcal{R}(T_1^{-1}) = H_0^2([a, b]).$$

Hence $(T_1^{-1})^{-1} = T_1$, and therefore T_1 is self-adjoint.

Let us now consider a Galerkin discretisation of the PDE

$$\ell u = -u'' + u'.$$

That is, we fix a discrete subspace H_n such that

$$\dim H_n = n,$$

and construct the sequence of discrete operators $\{T_n\}_{n \in \mathbb{N}}$,

$$T_n : H_n \rightarrow H_n, \quad (T_n)_{ij} = (\ell \phi_i, \phi_j),$$

where $\{\phi_i\}_{i=1}^n$ forms a basis of H_n .

In particular, we will focus our attention on the conforming setting, i.e.

$$H_n \subset H$$

for every $n \in \mathbb{N}$, and we will ask that for every $x \in H$, the operation ℓx makes sense at least from a distributional point of view.

Definition 0.20. Given two differential equations represented by their actions ℓh and $\widehat{\ell h}$, we say that they are formal L^2 adjoints on H of one another if

$$(\ell h, \widehat{h})_{L^2} = (h, \widehat{\ell h})_{L^2}$$

for every $\widehat{h} \in H$.

We observe that the sequence of operators $\{T_n^T\}_{n \in \mathbb{N}}$ can be defined in terms of the formal adjoint. Indeed,

$$(T_n)^T_{ij} = (\ell \phi_j, \phi_i) = (\widehat{\ell} \phi_i, \phi_j).$$

Hence, given a sequence of Galerkin discretisations of ℓ , their transposes $\{T_n^T\}_{n \in \mathbb{N}}$ are a sequence of Galerkin discretisations of the formal L^2 adjoint on H of ℓ .

We already discussed the difference between the Banach and Hilbert adjoints. In the following example, we clarify the difference between the formal adjoint and the Hilbert adjoint.

Example 0.21. Let us consider the ODE

$$\ell h = -h''$$

together with the boundary conditions

$$h(a) = h(b) = 0.$$

Integrating by parts, we can compute $(\ell h, \widehat{h})$:

$$\begin{aligned} (\ell h, \widehat{h})_{L^2} &= \int_a^b -h'' \widehat{h} \\ &= \int_a^b h' \widehat{h}' - h' \widehat{h} \Big|_a^b \\ &= - \int_a^b h \widehat{h}'' - h \widehat{h}' \Big|_a^b \\ &= - \int_a^b h \widehat{h}'' . \end{aligned}$$

The boundary terms vanish because $\widehat{h} \in H$ gives $\widehat{h}(a) = \widehat{h}(b) = 0$ and because $h \in H$.

This suggests that ℓ is its own formal L^2 adjoint, i.e.

$$\widehat{\ell} = \ell.$$

At first, this observation agrees with the fact that the solution operator we discussed in a previous remark is self-adjoint.

Yet we need to proceed with caution. In fact, for an operator to be self-adjoint, it needs to be an endomorphism. For this reason, we decide to consider the operator

$$T : L^2([a, b]) \rightarrow L^2([a, b]), \quad \text{dom}(T) = H_0^2([a, b]).$$

We are now in the setting of unbounded operators, which is well beyond the topic of these notes, so we will only make some comments. While the reader might think that T is the operator associated with ℓ and T' is the operator associated with $\widehat{\ell}$, and thus $\ell = \widehat{\ell}$ implies that T is self-adjoint, this is not really the case. In fact, while $\ell = \widehat{\ell}$ implies the symmetry of the operator T , we have no guarantee that

$$\text{dom}(T) \supset \text{dom}(T^*),$$

never mind the particular case of the Laplacian, where it is possible to construct a self-adjoint extension of T .

Not only is T' not the operator associated to $\widehat{\ell}$, but moreover, given a Galerkin discretisation $\{T_n\} : H \rightarrow H$, it is not true that $\{T_n^T\}_{n \in \mathbb{N}}$ is a Galerkin discretisation of T' .

If we consider as operator the one associated with the differential equation

$$\ell h = h'$$

with boundary conditions

$$h(a) = 0,$$

i.e.

$$T : L^2([a, b]) \rightarrow L^2([a, b]), \quad \text{dom}(T) = \{h \in H^1([a, b]) : h(a) = 0\},$$

then

$$\widehat{\ell} h = -h',$$

which shows that a transpose of a Galerkin discretisation is not discretising the adjoint operator, since the adjoint operator in this case has

$$\text{dom}(T') = \{h \in H^1([a, b]) : h(a) = 0\}.$$

A similar issue arises when discussing normality. Let us consider the operator associated with the differential equation

$$\ell h = h'$$

with boundary conditions

$$h(a) = h(b) = 0,$$

i.e.

$$T : L^2([a, b]) \rightarrow L^2([a, b]), \quad \text{dom}(T) = H_0^1([a, b]).$$

While we might think that the operator T is normal because

$$\ell \widehat{\ell} = -u'' = \widehat{\ell} \ell,$$

this is not the case. In fact,

$$\text{dom}(T^*) = H^1([a, b]),$$

and thus

$$\text{dom}(TT^*) \neq \text{dom}(T^*T),$$

proving that T is not a normal operator. See Kato, Example X.3.25.

Missing: a discussion on finite-rank operators.